THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The square of the distance between $(1, 4)$ and $(-3, 2)$ is: 1

(a) 17 **(b)** 25 **(c)** 20 **(d)** 22

2. If $x = 2$ is a zero of $7x + m$, then $m =$ 1

(a) -15 **(b)** -13 **(c)** -14 **(d)** -12

3. Which of the following has a terminating decimal expansion? 1

(a) $\frac{5}{18}$ **(b)** $\frac{14}{21}$ **(c)** $\frac{3}{7}$ **(d)** $\frac{39}{65}$

4. $(7x + 4y)^2 =$ 1

(a) $49x^2 - 56xy + 16y^2$ **(b)** $49x^2 + 16y^2$ **(c)** $49x^2 + 28xy + 16y^2$ **(d)** $49x^2 + 56xy + 16y^2$

5. The length of a chord of a circle of radius 5 cm lying 4 cm from the centre is (in cm): 1

(a) 6 **(b)** 5 **(c)** 8 **(d)** 7

6. The length of the arc of a sector of angle 90° in a circle of radius 14 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 23 **(b)** 24 **(c)** 22 **(d)** 21

7. Two dice are rolled together. The probability that the sum is 7 is: 1

(a) $\frac{1}{7}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{3}$

8. In an AP, $t_1 = 7$ and $t_{11} = 47$. The common difference is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

9. The converse of "If a number is divisible by 6, then it is divisible by 3" is false because: 1

(a) two angles of 60° in an equilateral triangle are equal but not vertically opposite

(b) $x = -3$ gives $x^2 = 9$ but $x \neq 3$

(c) 9 is divisible by 3 but not by 6

(d) a kite that is not a rhombus has perpendicular diagonals

10. The average of 12 numbers is 89. When a number 63 is included, the average of all 13 numbers becomes: 1

(a) 88 **(b)** 89 **(c)** 87 **(d)** 86

11. Using Euclid's subtraction algorithm for gcd$(360, 156)$, the number of subtraction steps needed is: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 11

12. The angles of a quadrilateral are in the ratio $5:3:3:1$. The largest angle is: 1

(a) 140° **(b)** 170° **(c)** 150° **(d)** 160°

13. The slope of the line $2x + 2y = 10$ is: 1

(a) -2 **(b)** 1 **(c)** 0 **(d)** -1

14. The radius of a sphere is multiplied by 6. Its surface area becomes ___ times: 1

(a) 12 **(b)** 216 **(c)** 36 **(d)** 6

15. The midpoint of $(3, 2)$ and $(6, 7)$ is: 1

(a) $(4.5,\ 4.5)$ **(b)** $(4.5,\ 5.5)$ **(c)** $(5,\ 4.5)$ **(d)** $(-1.5,\ -2.5)$

16. The slope of the line through $(1, 3)$ and $(3, 9)$ is: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 4

17. The number of decimal places in the decimal expansion of $\frac{7}{400}$ is: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

18. $(2x + 3)(3x + 5) =$ 1

(a) $6x^2 + 19x - 15$ **(b)** $6x^2 + x + 15$ **(c)** $6x^2 + 20x + 15$ **(d)** $6x^2 + 19x + 15$

19. A chord of length 8 cm lies 3 cm from the centre of a circle. The radius (in cm) is: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 7

20. A triangle has area 84 cm^2 and one side 14 cm. The altitude on this side (cm) is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 14

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the distance between $(1, 1)$ and $(4, 5)$. 2

22. Find the slope of the line through $(2, 4)$ and $(5, 13)$. 2

23. Express $0.8\overline{3}$ as a fraction. 2

24. Expand $(2x + 5)(3x + 1)$. 2

25. Find the length of the chord of a circle of radius 20 cm that is 16 cm from the centre. 2

26. A triangle has sides 25 cm, 25 cm and 14 cm. Find its area and the altitude on the side 14 cm. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Show that $A(1, 1)$, $B(4, 5)$, $C(10, 13)$ are collinear. 3

28. The line $y = mx + 12$ passes through $(2, 18)$. Find m, then the x-intercept of the line. 3

29. Express $0.\overline{027}$ as a fraction in lowest terms. 3

30. If $x + \frac{1}{x} = 6$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

31. Two parallel chords of lengths 16 cm and 30 cm lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. 3

OR

Find the area of the cyclic quadrilateral with sides 4 cm, 8 cm, 11 cm and 17 cm. 3

32. A die is rolled. Find the probability of getting 3

(i) a number less than 3,

(ii) a number greater than 4,

(iii) a number less than 3 or a number greater than 4.

OR

Find the number of terms of the AP $8, 14, 20, \ldots$ whose last term is 152, and its sum. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. The vertices of a rhombus are $A(-2, 2)$, $B(1, -2)$, $C(4, 2)$, $D(1, 6)$. Show that all four sides are equal and the diagonals are unequal, so that it is a rhombus and not a square. 5

34. A linear polynomial $p(x)$ has $p(1) = 8$ and $p(3) = 16$. Find $p(x)$, its zero and $p(5)$. 5

35. Express $1.1\overline{6}$ as $\frac{p}{q}$ and verify by long division of p by q. 5

36. If $x + y = 12$ and $x - y = 4$, find $x^2 - y^2$, xy and $x^2 + y^2$. 5

OR

A chord AB of a circle with centre O is 24 cm long and the radius is 13 cm. Find the distance of the chord from O and the area of triangle OAB. 5

37. A rectangular plot 40 m $\times$ 30 m has a circular pond of radius 7 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹6 per m^2. 5

OR

Two dice are rolled together. Find the probability of 5

(i) a sum of 5,

(ii) a sum of 9,

(iii) a sum of 5 or 9.

38. A ladder has 11 rungs. The bottom rung is 120 cm long and each rung above is 4 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

OR

Prove that the sum of 3 consecutive even numbers starting at $2n$ is a multiple of 6. Check with $n = 4$. 5

END OF QUESTION PAPER Total: 80 Marks

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